

Generic Rank-Three Failure of Cauchy-Dual Subnormality

A full-dimensional partial answer to the finite-support characterization problem

Candidate partial result for arXiv:2309.03588

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Status: candidate substantial partial result, likely valid. For measures supported at exactly three points, the packet proves that the subnormal locus is contained in a Lebesgue-null, nowhere-dense real-analytic exceptional set. It also proves non-subnormality throughout a full neighborhood of every equal-mass equilateral measure. The complete classification on the exceptional set remains open.

1 Source problem and claimed contribution

Khasnis and Sholapurkar state the finite-support Cauchy-dual problem in Section 1.2, Problem 1, on source PDF p. 5 [1]:

$$B_n(T) \leq 0 \text{ for all integers } n \geq 1.$$

Now we are ready to state and describe the problem that has been tackled in this article.

1.2 The Problem

We begin our discussion with the fact [1, Theorem 6.1] which states that for a cyclic, analytic 2-isometry T in $\mathcal{B}(\mathcal{H})$, the rank of $T^*T - I$ is finite if and only if there exist a finitely supported measure μ on the unit circle $\mathbb{T}$ such that T is unitarily equivalent to a multiplication by the coordinate function z on $D(\mu)$. In fact, if the rank of $T^*T - I$ is a positive integer k , then the corresponding measure μ is supported at exactly k points of unit circle $\mathbb{T}$. In view of this result, a construction of a cyclic, analytic 2-isometry reduces to choosing finitely many points on the unit circle and looking at the multiplication operator M_z on the Dirichlet space $D(\mu)$, where μ is supported at the chosen points. Further, it is asserted that such a space $D(\mu)$ coincides with the de Branges-Rovnyak space $H(B)$ for some B , with the equality of norms.

We now state the problem under consideration:

Problem 1. *Characterize finitely supported, positive, Borel measures μ on unit circle $\mathbb{T}$ such that the Cauchy dual of M_z on $D(\mu)$ is subnormal.*

A solution to the problem in the case when μ is supported at a single point is given by the following result (see [32, Corollary 3.6] and [1, Corollary 5.4]): Let $\lambda \in \mathbb{T}$ and $\gamma > 0$. The Cauchy dual of M_z on the Dirichlet-type space $D(\gamma\delta_\lambda)$ is a subnormal contraction. Further, it is proved in [1, Theorem 2.4] that for a positive, Borel measure μ supported at any two antipodal points on unit circle $\mathbb{T}$, the Cauchy dual M'_z of M_z is subnormal. In view of these two results, it is natural to investigate the case of a measure supported at two non-antipodal points and address the problem in that case.

Here, we consider the measures supported at two points of unit circle $\mathbb{T}$ such that the angle between them is 90° . Indeed, the main result of this paper is as given below.

Theorem 1.1. *Let ζ_1 and ζ_2 be any two points on the unit circle $\mathbb{T}$ such that the angle between them is 90° (considered as vectors in $\mathbb{R}^2$). If $\mu = \delta_{\zeta_1} + \delta_{\zeta_2}$ is a measure on unit circle $\mathbb{T}$ then the Cauchy dual of M_z on the Dirichlet-type space $D(\mu)$ is not subnormal.*

The rest of the paper is devoted to the proof of this result. The nature of the proof is computational. Although, the computational details are not included in the proof, the same can be verified by an application of the SageMath code provided in the appendix.

Exact transcription. “Characterize finitely supported, positive, Borel measures μ on unit circle $\mathbb{T}$ such that the Cauchy dual of M_z on $D(\mu)$ is subnormal.”

For ordered three-point measures, normalize rotation and cyclic order by writing

$$\mu_\lambda = c_1\delta_1 + c_2\delta_{e^{i\theta_2}} + c_3\delta_{e^{i\theta_3}}, \quad \lambda \in \Omega, \tag{1}$$

where

$$\Omega = \{(\theta_2, \theta_3, c_1, c_2, c_3) : 0 < \theta_2 < \theta_3 < 2\pi, c_j > 0\}.$$

This connected open set is a full parameter chart for rank-three measures, up to rotation and relabeling.

Theorem 1 (Generic rank-three non-subnormality). *There are two nonzero real-analytic functions $\mathcal{D}, \mathcal{R}$ on Ω such that, if the Cauchy dual of $\mathcal{M}_z$ on $D(\mu_\lambda)$ is subnormal, then*

$$\mathcal{D}(\lambda)\mathcal{R}(\lambda) = 0.$$

Consequently the subnormal rank-three measures form a Lebesgue-null and nowhere-dense subset of every chart (1). Equivalently, an open dense set of rank-three measures, and almost every such measure in the natural angle–mass coordinates, has a non-subnormal Cauchy dual.

Corollary 2 (Robust equilateral obstruction). *Fix $c > 0$, $\xi \in \mathbb{T}$, and $\omega = e^{2\pi i/3}$. There is an $\varepsilon > 0$ such that, whenever the atoms can be labeled so that*

$$|\zeta_j - \xi\omega^j| < \varepsilon, \quad |c_j - c| < \varepsilon \quad (j = 0, 1, 2),$$

the Cauchy dual on $D(\sum_{j=0}^2 c_j \delta_{\zeta_j})$ is not subnormal. Thus the known equilateral example lies inside a full-dimensional open set of arbitrary support triples and arbitrary positive masses.

2 Rational model and the forbidden-atom criterion

For a finite positive measure μ on $\mathbb{T}$, $D(\mu)$ denotes the corresponding Dirichlet-type space, and $\mathcal{M}_z f = zf$ is a cyclic analytic 2-isometry. Its Cauchy dual is $\mathcal{M}'_z = \mathcal{M}_z(\mathcal{M}_z^* \mathcal{M}_z)^{-1}$.

The rational de Branges–Rovnyak model of Chavan–Ghara–Reza and Costara [2, 3] gives, for every $\lambda \in \Omega$, a monic outer polynomial

$$q_\lambda(z) = \prod_{r=1}^3 (z - \alpha_r), \quad |\alpha_r| > 1,$$

and a row Schur function

$$B = (p_1/q_\lambda, p_2/q_\lambda, p_3/q_\lambda)$$

such that $D(\mu_\lambda) = \mathcal{H}(B)$ isometrically. Put

$$G_\lambda(z, w) = \sum_{j=1}^3 p_j(z) \overline{p_j(w)}. \quad (2)$$

The polynomial G_λ is intrinsic even though a matrix factor producing the individual p_j is not unique.

When the roots of q_λ are simple, set

$$\kappa_{rt} = \frac{G_\lambda(\alpha_r, \alpha_t)}{\alpha_r^2 \overline{\alpha_t}^2 q'_\lambda(\alpha_r) \overline{q'_\lambda(\alpha_t)}}. \quad (3)$$

Corollary 4.1 and Remark 4.2 of [2] give the following necessary condition: if $\mathcal{M}'_z$ is subnormal, then

$$\nu_\lambda = \sum_{r,t=1}^3 \kappa_{rt} \delta_{1/(\alpha_r \overline{\alpha_t})} \quad (4)$$

is a positive measure supported on $[0, 1]$. Equal atom locations in (4) must be grouped before their coefficients are tested. In particular, every grouped coefficient outside $[0, 1]$ must vanish.

The next observation sharpens Corollaries 2.3 and 4.3 of [2] by removing every hypothesis on the pole products.

Lemma 3 (Simple-root orthogonality criterion). *If the roots $\alpha_1, \alpha_2, \alpha_3$ are distinct, then the Cauchy dual is subnormal if and only if*

$$G_\lambda(\alpha_r, \alpha_t) = 0 \quad (r \neq t). \quad (5)$$

Proof. Put $u_r = 1/\alpha_r$ and let c_{rt} denote the quantity in (3). Theorem 2.1 of [2] says that subnormality makes, for every integer $\ell \geq 1$, the infinite matrix

$$\left(\sum_{r,t=1}^3 c_{rt} (1 - u_r \overline{u_t})^\ell u_r^m \overline{u_t}^n \right)_{m,n \geq 0}$$

formally positive semidefinite. Compress its leading 3×3 block and conjugate by the inverse of the Vandermonde matrix $(u_r^m)_{0 \leq m \leq 2, 1 \leq r \leq 3}$. This gives

$$C_\ell = (c_{rt} (1 - u_r \overline{u_t})^\ell)_{r,t=1}^3 \succeq 0.$$

If $G(\alpha_r, \alpha_t) \neq 0$, positivity of the numerator Gram kernel makes both corresponding diagonal values positive. The (r, t) principal minor of C_ℓ would therefore require

$$c_{rr} c_{tt} ((1 - |u_r|^2)(1 - |u_t|^2))^\ell \geq |c_{rt}|^2 |1 - u_r \overline{u_t}|^{2\ell}.$$

But

$$|1 - u_r \overline{u_t}|^2 - (1 - |u_r|^2)(1 - |u_t|^2) = |u_r - u_t|^2 > 0,$$

so the displayed inequality fails for all sufficiently large ℓ . Thus every cross value vanishes. Conversely, Corollary 2.3 of [2] proves subnormality when all cross values vanish. $\square$

Proof Intuition

There are two independent algebraic obstructions. The discriminant of the outer factor detects repeated roots. Away from that hypersurface, Lemma 3 forces all cross-Gram values to vanish, and one iterated resultant is the product of all nine Gram values $G(\alpha_r, \alpha_t)$. Neither analytic function is identically zero: the exact equilateral calculation witnesses both facts. Proper real-analytic zero sets have measure zero and empty interior.

The neighborhood result uses more information at the equilateral point. Its six off-diagonal residues form two clusters, and all three residues in either cluster have the same nonzero value. Under a small perturbation, even if atoms collide, no nonempty sum of nearby residues can cancel. Hence a forbidden atom remains outside $[0, 1]$.

3 Analytic dependence and the two obstructions

Lemma 4 (Analytic rational data). *The coefficients of q_λ and G_λ depend real-analytically on $\lambda \in \Omega$.*

Proof. Let $s_\lambda(z) = \prod_{j=1}^3 (z - \zeta_j)$. The outer factor is uniquely determined, together with $\gamma_\lambda > 0$, by

$$|s_\lambda(z)|^2 + \sum_{j=1}^3 c_j \left| \frac{s_\lambda(z)}{z - \zeta_j} \right|^2 = \gamma_\lambda |q_\lambda(z)|^2, \quad z \in \mathbb{T}. \quad (6)$$

The Laurent coefficients on the left are real-analytic in λ . The coefficient map $(\gamma, q) \mapsto \gamma |q|^2|_{\mathbb{T}}$, with q monic and zero-free on $\mathbb{D}$, has invertible linearization. Indeed, a kernel vector would give on $\mathbb{T}$

$$a + 2 \operatorname{Re}(h/q) = 0,$$

where $a \in \mathbb{R}$ and $\deg h < \deg q$. The analytic function h/q on $\mathbb{D}$ has constant real part, hence is constant; the degree restriction then forces $h = 0$, and also $a = 0$. The real-analytic implicit-function theorem and uniqueness of the outer factor give analytic dependence of q .

Costara's kernel formula expresses G using q , s , the rational functions

$$O = \gamma^{-1/2} e^{i\vartheta} s/q, \quad f_j = \frac{O}{O'(\zeta_j)(z - \zeta_j)},$$

and the inverse of the positive Gram matrix $(\langle f_i, f_j \rangle)_{i,j=1}^3$; see Theorem 6.1 and Remark 6.2 of [2]. Every denominator stays nonzero on Ω , and the phase normalization cancels in the intrinsic product (2). These finite algebraic operations and a matrix inverse preserve real-analytic dependence. $\square$

Define the outer-root discriminant

$$\mathcal{D}(\lambda) = \text{Disc}_z q_\lambda(z). \quad (7)$$

Thus $\mathcal{D} \neq 0$ exactly when the three outer roots are distinct.

Write $\widehat{G}_\lambda(z, y)$ for the polynomial obtained from $G_\lambda(z, w)$ by replacing $\bar{w}$ by y , and put $q_\lambda^c(y) = \overline{q_\lambda(\bar{y})}$. Define

$$\mathcal{R}(\lambda) = \text{Res}_z \left(q_\lambda(z), \text{Res}_y \left(q_\lambda^c(y), \widehat{G}_\lambda(z, y) \right) \right). \quad (8)$$

Both polynomials q_λ and q_λ^c are monic, so the elementary root formula for resultants gives

$$\mathcal{R}(\lambda) = \prod_{r,t=1}^3 G_\lambda(\alpha_r, \alpha_t) = \prod_{r=1}^3 G_\lambda(\alpha_r, \alpha_r) \prod_{r < t} |G_\lambda(\alpha_r, \alpha_t)|^2. \quad (9)$$

In particular $\mathcal{R}$ is real-valued. Lemma 4 shows that $\mathcal{D}$ and $\mathcal{R}$ are real-analytic on Ω .

4 The two nonzero witnesses

Lemma 5 (The resultant is nontrivial). *$\mathcal{R}$ is not identically zero on Ω .*

Proof. Fix $c > 0$ and take the equal-mass equilateral measure $c(\delta_1 + \delta_\omega + \delta_{\omega^2})$. Root-of-unity orthogonality in (6) gives

$$q(z) = z^3 - \rho, \quad \rho + \rho^{-1} = 2 + 9c, \quad \rho > 1.$$

The circulant Costara Gram matrix diagonalizes in the Fourier basis and yields

$$G(z, w) = P(z\bar{w}), \quad P(x) = \sum_{k=1}^3 A_k x^k, \quad A_k = \frac{3\rho(\rho-1)^2}{[3 + (k-1)(\rho-1)][3 + k(\rho-1)]} > 0. \quad (10)$$

This is the $N = 3$ specialization of the same rational-kernel calculation used in [4]; it also follows directly from Theorem 6.1 of [2].

Put $a = \rho^{1/3}$. The roots of q are $a, a\omega, a\omega^2$. The diagonal value $P(a^2)$ is positive. If $P(a^2\omega) = 0$, real coefficients also give $P(a^2\omega^2) = 0$. Hence $P(x)/x$ would be a multiple of

$$(x - a^2\omega)(x - a^2\omega^2) = x^2 + a^2x + a^4,$$

forcing $A_2/A_1 = a^{-2} = \rho^{-2/3}$. But (10) gives

$$\frac{A_2}{A_1} = \frac{3}{3 + 2(\rho - 1)} < \rho^{-2/3}.$$

For the strict inequality, write $a > 1$ and note that

$$1 + \frac{2(a^3 - 1)}{3} - a^2$$

vanishes at $a = 1$ and has derivative $2a(a - 1) > 0$. Thus both off-diagonal values are nonzero. Every factor in (9) is nonzero, so $\mathcal{R} \neq 0$ at this equilateral point. $\square$

Lemma 6 (The outer-root discriminant is nontrivial). *$\mathcal{D}$ is not identically zero on Ω .*

Proof. At the equal-mass equilateral point used above, $q(z) = z^3 - \rho$ with $\rho > 1$. Its three roots are distinct, so $\mathcal{D} \neq 0$ there. $\square$

5 Proofs of the main results

Proof of Theorem 1. Suppose $\mathcal{D}(\lambda) \neq 0$. Then the roots of q_λ are distinct. If the Cauchy dual were subnormal, Lemma 3 would force (5); then (9) would give $\mathcal{R}(\lambda) = 0$. Therefore every subnormal parameter belongs to

$$\{\mathcal{D} = 0\} \cup \{\mathcal{R} = 0\}.$$

By Lemmas 5 and 6, these are the zero sets of two nonzero real-analytic functions on the connected open set Ω . A proper real-analytic zero set has Lebesgue measure zero and empty interior [5]. Their finite union has the same properties, proving the theorem. $\square$

Proof of Corollary 2. Rotation is harmless, so take $\xi = 1$. At the equal-mass equilateral point, label $\alpha_r = a\omega^r$, $r = 0, 1, 2$. If $d = r - t \not\equiv 0 \pmod{3}$, then (3) and $q'(\alpha_r) = 3\alpha_r^2$ give the common residue

$$\kappa_{rt} = K_d := \frac{P(a^2\omega^d)}{9\rho^2 a^2\omega^d} \neq 0. \quad (11)$$

Thus the three ordered pairs with difference $d = 1$ have one common nonzero residue, and the three with difference $d = 2$ have another. Their atom locations are the two nonreal points $a^{-2}\omega^{-1}$ and $a^{-2}\omega^{-2}$.

The outer roots stay simple and can be labeled continuously under a small perturbation. Lemma 4 and (3) show that the six residues and their atom locations vary continuously. Choose disjoint neighborhoods U_1, U_2 of the two base locations, both disjoint from $[0, 1]$. Shrink the parameter neighborhood until every residue in cluster d differs from K_d by less than $|K_d|/2$.

Any collision group inside U_d is a nonempty sum of, say, m such residues. Its distance from mK_d is less than $m|K_d|/2$, so the grouped coefficient is nonzero. Hence the measure (4) has a nonzero atom outside $[0, 1]$, contradicting the necessary condition for subnormality. This holds for every sufficiently small perturbation, including parameters where some pole products collide. $\square$

6 Verification, limitations, and novelty audit

The accompanying script `code/verify_rank_three.py` independently constructs the degree-six self-inversive polynomial, selects the three outer roots, reconstructs Costara’s Gram matrix and the intrinsic polynomial G , and evaluates the nine pole products, six cross values, and six forbidden residues. It passes three cases: the equal-mass equilateral model, a fully nonsymmetric perturbation, and a small-mass generic witness. In the latter two cases all nine pole products are separated and every cross value is nonzero. These floating-point checks are sanity tests, not proof.

Adversarial verification focus. The most delicate points are (i) the real-analytic dependence in Lemma 4 and (ii) the Vandermonde compression and large- ℓ principal-minor argument in Lemma 3. The local stability proof separately handles pole-product collisions by grouping the residues exactly as required by Remark 4.2 of [2].

Scope limitation. The theorem does not classify the proper real-analytic exceptional set $\{\mathcal{DR} = 0\}$. Subnormal points, if any beyond the known boundary and lower-rank cases, must lie there, but membership in that set is not sufficient for subnormality. The full arbitrary-three-point conjecture and the characterization for four or more atoms remain open.

Bounded novelty audit. The run indexes and primary-source arXiv searches through 26 August 2026 used arXiv:2103.10059, 2309.03588, 2510.14004, and 2511.04565; the authors Khasnis, Phatak, Sholapurkar, Chavan, Ghara, and Reza; and the phrases “Cauchy dual subnormality,” “rank three,” “three distinct points,” “almost every,” “generic,” and “perturbation.” The search found the rational-symbol criterion, the original rank-two example, the later rank-two classification, and the equal-mass equilateral rank-three example [2, 1, 4]; it found no generic, full-measure, or perturbative rank-three theorem. Novelty is plausible, not certified.

Human-review recommendation. Send to an operator theorist familiar with rational de Branges–Rovnyak spaces. If the simple-root criterion and analytic-dependence lemma pass review,

the generic theorem is a checkable substantial partial answer; the exceptional set is then the natural target for a full classification.

References

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